STAT 486 Project 4

Model Comparisons For Crop Yield Studies

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Abstract

In this report, observational data from a crop yield experiment is utilized to understand the effects of the fertilizers and the plant densities on crop yield, as well as the blocking effect. Three linear regression models were constructed and compared, after which the best model was chosen based on the AIC value. Model B had the lowest AIC value of 173.1895. Both factors within the model were determined to be significant, with fertilizer 3 found to have a significant difference from fertilizers 1 and 2. There was a significant difference between low plant density and high plant density as well.

Introduction

Various aspects of the planting environment were observed to analyse the different effects upon crop yield. Collected variables are listed in table 1. The aim of this report is to assess different models and select the best one in order to identify the significant main effects.

Table 1: Variables

yield	Final crop yield (bushels per acre)
fertilizer	Type of fertilizer used (fertilizer type 1,2, or 3)
density	Planting density (1 =low density, 2 =high density)
block	Block in the field (1,2,3,4)

Three regression models were obtained, and titled models A, B, and C respectively. Model A utilizes two factors and their interaction effect as predictors, while Model B only utilizes two factors, and Model C contains all factors, including blocking effect.

Formula 1: (Model A) Crop Yield by Fertilizer, Density, Fertilizer and Density

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \epsilon_i, \ \epsilon_i \sim N(0, 1)$$

Formula 2: (Model B) Crop Yield by Fertilizer, Density

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon_i, \ \epsilon_i \sim N(0, 1)$$

Formula 3: (Model C) Crop Yield by Fertilizer, Density, Fertilizer and Density, Block

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_1 x_2 + \epsilon_i, \ \epsilon_i \sim N(0, 1)$$

Procedure

The dataset was uploaded into Rstudio, where it was examined. Prior to any operations, the three categorical variables were converted into factors, with the density variable being converted into an ordered factor. The value “1” was attributed to low density, and “2” was attributed to high density. The column sums for each variable and their factor levels were calculated. These values are displayed in the tables below.

Table 2: Bushels per acre for respective density levels

Low Density	High Density
8485.656	*8507.830*

Table 3: Bushels per acre for respective fertilizer type

Type 1	Type 2	Type 3
5656.225	*5661.863*	*5675.397*

Table 4: Bushels per acre for respective block

Block 1	Block 2	Block 3	Block 4
4244.553	*4255.605*	*4241.103*	*4252.225*

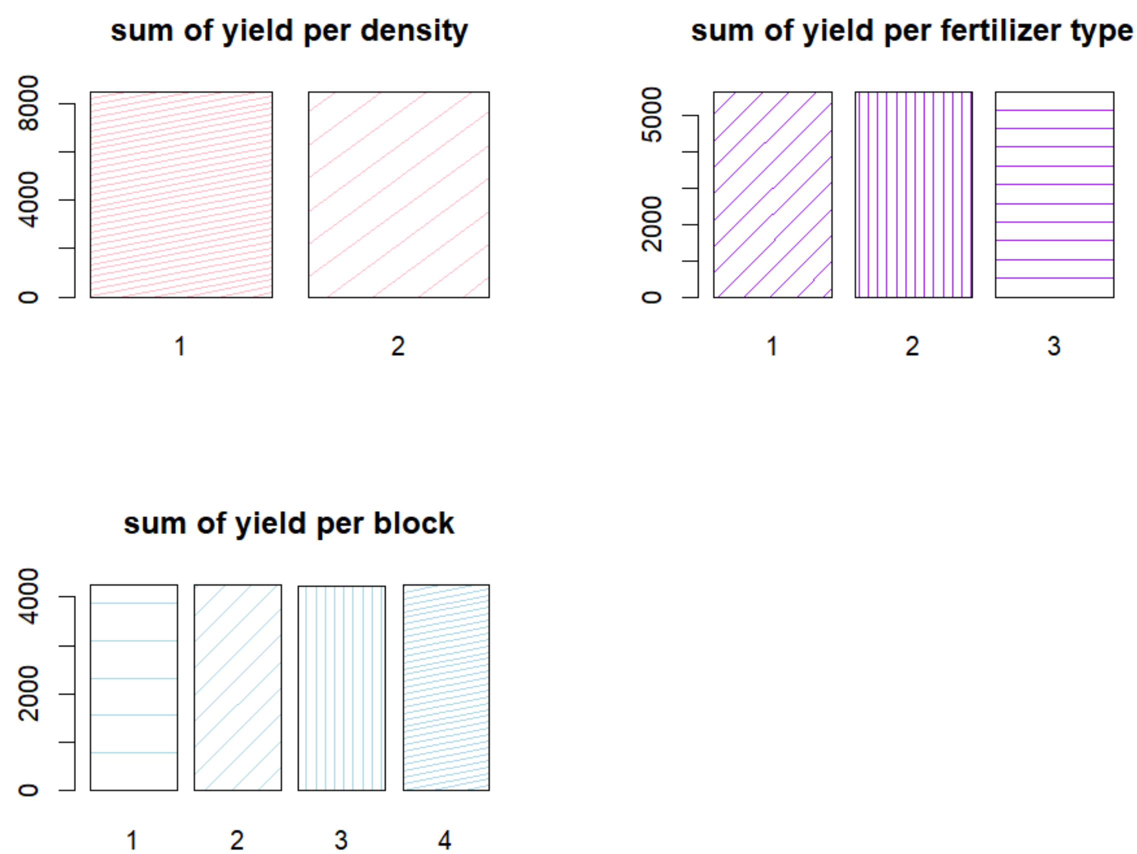
From a cursory look at the numbers, each partition of data appears to be evenly split, with minimal differences between the various levels. Similarly, when observing the 5-number summary for the crop yield variable, there is very little deviation from the mean for all values. The observed numbers are very consistent. The summary is detailed in table 5.

Table 5: Crop yield summary statistics

Min.	1st Qu.	Median	Mean	3rd Qu.	Max
175.4	*176.5*	*177.1*	*177.0*	*177.4*	*179.1*

To further visualize this observation, a barplot of each set of counts was obtained. This plot is pictured in figure 1. Visualized in the plot, the same observation of consistency can be made.

Figure 1: Bar plot of crop yield counts



Before the AIC values for the various models are calculated, the different effects within the models are observed, and tests are conducted to determine significance. First, model A is assessed to determine whether the interaction effect between fertilizer and density is significant. The ANOVA table is observed for the associated F-statistic and p-value for the interaction effect. With an F-statistic of 0.6346 and a p-value of 0.5325, the interaction is deemed to not be significant for the response variable. Next, to determine whether the blocking effect is significant, the ANOVA F-test was performed for models A and C. From the results, the F-statistic value 0.7166 was obtained, as well as the p-value 0.4913. Due to this large p-value, blocking effect was determined to be insignificant for the response variable. When examining the ANOVA table for model B, both factors were found to be highly significant. Variable fertilizer was found to have a p-value of 0.0002533, with the respective F-statistic of 9.0731. Variable density was found to have a p-value of 0.0001741, and an F-statistic of 15.3162. These p-values are very small, and so these variables will be noted for post-hoc tests.

The AIC values of each model were calculated, to determine the model with the best fit. The obtained values are displayed in table 6.

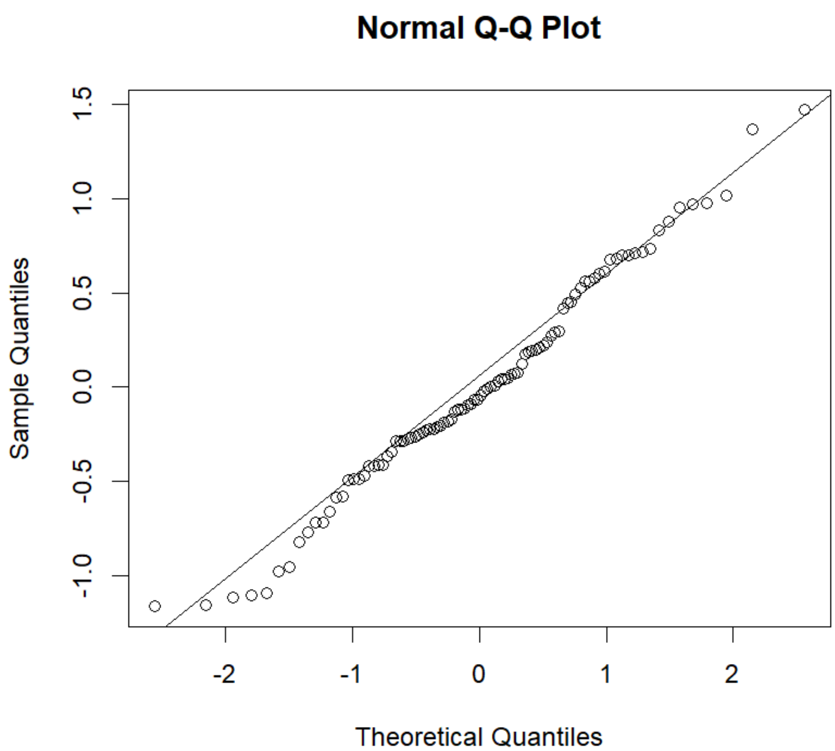
Table 6: AIC values

Model A	Model B	Model C
175.8451	*173.1895*	*178.2943*

Notably, model B was associated with the smallest AIC value, or the best fit. However, it is important to note that the AIC method prefers models with fewer variables, and model B is the simplest model of the selections. The most complex model (C) has the highest AIC value, to compare. Ultimately, the difference between the three values is not notably large. Model B is chosen to proceed with the analysis.

Before performing the analysis on model B, the model assumptions are checked. Additionally, equal variance of the data is checked in order to later perform Tukey's test.

Figure 2: Normality of Residuals



Though the residuals do not quite seem to follow the reference line very closely, the analysis will proceed with the assumption of normality due to the robustness of the ANOVA test.

Levene's test for homogeneity of variance is conducted for each factor in the data set. The results of each test are displayed in table 7. Levene's test does not require normality of data.

Table 7: Levene's test

	Yield~Fertilizer	Yield~Density	Yield~Block
F-statistic	*0.8472*	*0.152*	*0.1388*
P-value	*0.4319*	*0.6975*	*0.9366*

The results of each test conclude that all variances can be assumed to be equal (due to the high p-values found in each iteration).

Previously, the main effects were determined to be the fertilizer and density variables, without interactions. The coefficients for these variables are displayed in formula 4.

Formula 4: Model B

$$Y = 176.757 + 0.1762x_1 + 0.5991x_1 + 0.3267x_2 + \epsilon_i$$

Lastly, in order to determine differences between the main effects, The Tukey test was performed. Significant differences were found between fertilizer type 3, and the first two fertilizer types. Types 1 and 2 were not found to have significant differences between each other, while both had significant differences from type 3. The most significant difference was found between type 3 and type 1, with a p-adj value of 0.0002219. The mean difference between these types was 0.5991256.

Variable density also displayed significant differences between high and low density observations, with a mean difference of 0.461956 and a p-adj value of 0.0001741. The plot for this test is displayed in figure 3.

Figure 3: Intervals for Tukey test

Conclusion

After concluding the analysis of the different aspects of this dataset, fertilizer and plant density were both found to have significant effects on crop yield. Beyond this, there were notable differences within the type of fertilizer used, as well as high or low density. The largest difference was found between type 1 and type 3 fertilizer, with a p-adj value of 0.0002219 and a mean difference of 0.5991256. The model with the lowest AIC value of *173.1895* was model B, containing only two factors for prediction.

Appendix (R code)

```
# STAT486 PROJECT 4
# HEBA SYED

setwd("C:/Users/heb/Documents/STAT")

crops<-read.csv("CropData.csv")
attach(crops)

str(crops)

tapply(yield, density, sum)
#        1        2
# 8485.656 8507.830
tapply(yield, fertilizer, sum)
# 1        2        3
# 5656.225 5661.863 5675.397
tapply(yield, block, sum)
# 1        2        3        4
# 4244.553 4255.605 4241.103 4252.225

summary(yield)

par(mfrow=c(2,2))
barplot(tapply(yield, density, sum), main = "sum of yield per
density", density = c(30,7), angle = c(11,36), col = "pink")
barplot(tapply(yield, fertilizer, sum), main = "sum of yield per
fertilizer type", density = c(10,20, 10), angle = c(45,90,0), col =
"purple")
```

```
barplot(tapply(yield, block, sum), main = "sum of yield per block",
density=c(5,10,20,30,7) , angle=c(0,45,90,11,36), col = "lightblue")
par(mfrow=c(1,1))

crops$density<-ordered(as.factor(density))
crops$block<-as.factor(block)
crops$fertilizer<-as.factor(fertilizer)

modelA<-lm(yield~fertilizer+density+fertilizer*density, data = crops)
summary(modelA)
anova(modelA) # mse1: 0.3371
# interaction effect is not significant (0.5325)

modelB<-lm(yield~fertilizer+density, data = crops)
summary(modelB)



modelC<-lm(yield~fertilizer+density+fertilizer*density+block, data =
crops)

anova(modelA, modelC, test="F")
# Model 1: yield ~ fertilizer + density + fertilizer * density
# Model 2: yield ~ fertilizer + density + fertilizer * density + block
# Res.Df    RSS Df Sum of Sq      F Pr(>F)
# 1     90 30.337
# 2     88 29.851  2   0.48614 0.7166 0.4913
# blocking effect not significant

AIC(modelA) # 175.8451
AIC(modelB) # 173.1895 - lowest AIC value
```

```
AIC(modelC) # 178.2943



summary(modelB)
anova(modelB)
# both are significant,

# assumption check
qqnorm(modelB$residuals)
qqline(modelB$residuals)

par(mfrow=c(2,2))
plot(modelB)

library(car)
leveneTest(yield~fertilizer, data = crops)
leveneTest(yield~density, data = crops)
leveneTest(yield~block, data = crops)
# equal var

tk<-TukeyHSD(aov(modelB), ordered = TRUE, conf.level = 0.95)
tk
par(mfrow=c(1,2))
plot(tk, las=1, col= "green")
TukeyHSD(aov(modelA), ordered = TRUE)
```